Daniel Brosch

Curriculum Vitae

Research Interests

Semidefinite and polynomial optimization; symmetry reduction and representation theory; algorithms for conic optimization; computer-assisted proofs in extremal combinatorics and discrete geometry.

Academic and Research Appointments

Sep 2026–present **Researcher / MSCA Postdoctoral Fellow**, *Department of Mathematics and Statistics and the Lie-Størmer Center, UiT The Arctic University of Norway*, Tromsø, Norway
PRISM, hosted by Cordian Riener.

Dec 2022–present **Assistant Professor (non-tenure-track)**, *Department of Mathematics, University of Klagenfurt*, Austria
Six-year appointment (on leave for the MSCA Fellowship).

Sep–Nov 2022 **Senior Scientist**, *Department of Mathematics, University of Klagenfurt*, Austria

Nov 2018–Jul 2022 **Doctoral researcher**, *Tilburg University*, the Netherlands

Jan–Mar 2020 **Researcher**, *Centrum Wiskunde & Informatica*, Amsterdam, the Netherlands

Oct–Dec 2019 **Researcher**, *Ortec*, Zoetermeer, the Netherlands

Education

2018–2022 **PhD in Mathematics**, *Tilburg University*, the Netherlands
Thesis: *Symmetry Reduction in Convex Optimization with Applications in Combinatorics*; defended 19 October 2022. Supervisors: Etienne de Klerk and Monique Laurent. Early-stage researcher in the Marie Skłodowska-Curie Innovative Training Network MINOA.

2017–2018 **Mathematics MSc**, *University of Cologne*, *cum laude*
Thesis: *Semidefinite Bounds for Unequal Error Protection Codes*, under supervision of *Frank Vallentin*.

2015–2017 **Mathematics BSc**, *University of Cologne*, *cum laude*
Thesis: *The Banach-Tarski Paradox*, under supervision of *Alexander Lytchak*.

2012–2015 **Early university studies**, *University of Cologne*, Germany
Schülerinnen und Schüler an der Universität: university courses taken alongside secondary school.

2007–2015 **Abitur**, *Otto-Hahn-Gymnasium*, Monheim am Rhein
Mathematics, physics, Latin and philosophy.

Department of Mathematics and Statistics
UiT The Arctic University of Norway, Tromsø, Norway
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Research Funding and Awards

2026–2029 **MSCA Postdoctoral Fellowship: PRISM**, *Marie Skłodowska-Curie Actions, 2025 call*, EUR 251,578.54

Graph Profiles via Invariants, Sums-of-squares and Moment methods. Started 1 September 2026; 24 funded research months. Host: Cordian Riener, UiT The Arctic University of Norway. Planned four-month secondment with Oleg Pikhurko, University of Warwick.
The project develops symbolic and semidefinite methods for certifying boundaries of polynomial images and graph profiles, combining polynomial optimization, real algebraic geometry, representation theory and extremal combinatorics.

2026 **Abbe Grant**, *My attendance of this years Heidelberg Laureate Forum was supported by the Carl-Zeiss-Stiftung*

2025 **ÖFG Forschungsförderprogramm „Internationale Kommunikation“**, *ÖFG*, EUR 350

2025 **Lorentz Center travel grant**, *Lorentz Center, Leiden*, EUR 600

2023 **Young Scientist Mentoring**, *University of Klagenfurt*, EUR 3,100
Grant for high-performing early career researchers (competitive selection procedure) to visit or invite their mentor. Mentor: *Greg Blekherman, Georgia Tech*

2021 **Talent Grant**, *Tilburg University*
Internal grant for a 9-month contract extension during my PhD (competitive selection procedure).

Publications

Journal articles

2026 **The Augmented Mixing Method: Computing High-Accuracy Primal-Dual Solutions to Large-Scale SDPs via Column Updates**, Optimization Methods and Software *41(3), 754–796*, with Jan Schwiddessen and Angelika Wiegele
We developed a novel SDP-solver based the Burer-Monteiro factorization, modified to only update one column of the factors at a time. Trough a novel update rule and direct handling of inequalities, we are able to solve SDPs with tens of millions of inequalities to high precision, which is out of reach of any other solver in the literature.
`https://doi.org/10.1080/10556788.2026.2683647`

2026 **Getting to the Root of the Problem: Sums of Squares for Limits of Trees**, Mathematical Programming *218, 505–561*, with Diane Puges
Trees can be considered dense objects when we only consider the leaves of the trees to be its vertices. This leads to a natural theory of limits of trees, first considered by Czabarga, Székely and Wagner, allowing us to ask questions of the form "How many copies of a small tree can an infinite tree contain?". We define the flag algebra of binary trees in this setting, and use it to recover all existing bounds on the inducilibities of trees with up to 10 leaves, and compute hundreds of completely new bounds, including new sharp bounds. Finally, we compute the first outer approximations of profiles of trees, which encode the possible simultaneous densities of multiple small trees in an infinite tree. Published online in 2025.
`https://doi.org/10.1007/s10107-025-02305-1`

2024 **New lower bounds on crossing numbers of $K_{m,n}$ from semidefinite programming**, Mathematical Programming *207, 693–715*, with Sven Polak
We develop representation theoretic tools to simplify SDP bounds for the crossing number of complete bipartite graphs, improving bounds both in the finite case and in the limit. We also introduce a new, slightly weaker, but computationally more efficient bound for the crossing number of $K_{m,n}$, allowing us to compute bounds for bigger parameters m and n. Published online in 2023.
`https://doi.org/10.1007/s10107-023-02028-1`

2022 **Jordan symmetry reduction for conic optimization over the doubly nonnegative cone: theory and software**, Optimization Methods and Software *37(6), 1987–2006*, with Etienne de Klerk

We extend the Jordan Reduction, a method to exploit symmetries based on Jordan algebras, to the doubly nonnegative cone, and provide a Julia implementation of the algorithm.
`https://doi.org/10.1080/10556788.2021.2022146`

2022 **Minimum energy configurations on a toric lattice as a quadratic assignment problem**, Discrete Optimization *44, article 100612*, with Etienne de Klerk

We bound the potential energy of charged particles on an infinite, periodic grid from below, using semidefinite programming and symmetry reduction based on the Jordan Reduction method.
`https://doi.org/10.1016/j.disopt.2020.100612`

2021 **Optimizing Hypergraph-Based Polynomials Modeling Job-Occupancy in Queuing with Redundancy Scheduling**, SIAM Journal on Optimization *31(3), 2227–2254*, with Monique Laurent and Andries Steenkamp

We show that a family of highly symmetric polynomials is convex, thus (partially) solving a conjecture from queueing theory with redundancy scheduling. To do so, we exploit the symmetries of the Hessians of the polynomials algebraically.
`https://doi.org/10.1137/20M1369592`

Preprints and submitted manuscripts

2025 **Lower and Upper Bounds for Small Canonical and Ordered Ramsey Numbers**, with Bernard Lidický, Sydney Miyasaki and Diane Puges

We compute new bounds on three variants of Ramsey numbers of small graphs. They are generalizations for ordered graphs, graphs with infinitely many edge-colors, and both. We use integer programming for lower and upper bounds, flag algebras for upper bounds and obtaining information on potential constructions, and heuristic search algorithms to find them. We recovered many known Ramsey numbers, and found multiple new Ramsey numbers.
`https://arxiv.org/abs/2511.04364`

2025 **Lattice paths with dynamic boundary**, *Submitted*, with Andrei Asinowski and Sarah Selkirk

We consider directed lattice paths where the occurrence of a pattern shifts the boundary upwards or downwards by a certain quantity. We show that their generating functions are algebraic by constructing one-counter automatons which describe them, and that they are ultimately equivalent to coloured pattern-avoiding directed lattice paths. We further demonstrate that lattice paths with dynamic boundary exhibit interesting links with other combinatorial structures. As an example, we establish a connection between one such model and shifted Young tableaux: we provide a bijection, and find a bivariate generating function by the vectorial kernel method.

Research Software and Digital Skills

2025– **AugmentedMixing.jl**

Julia implementation of the Augmented Mixing Method for semidefinite programs.
`https://github.com/jschwiddessen/AugmentedMixing.jl`

2021– **FlagSOS.jl**

Extensible Julia package for fully symmetry-reduced flag sums-of-squares problems for different combinatorial structures.
`https://github.com/DanielBrosch/FlagSOS.jl`

2021 **SDPSymmetryReduction.jl**

Julia package for automatic symmetry reduction of SDPs using Jordan reduction.
`https://github.com/DanielBrosch/SDPSymmetryReduction.jl`

Programming Julia, C/C++, Python, Java and MATLAB; additional experience with SageMath, JavaScript and C#.

Teaching tools Interactive mathematical visualizations and quizzes; some experience with H5P, Tweedback and Slido.

Supervision

Doctoral students

2025– **Johannes Schmucker**, *SynMaxCut: Synergies for Exact Solutions to the Maximum Cut Problem*, Klagenfurt
Joint supervision with Angelika Wiegele

2022–2026 **Jan Schwiddessen**, *Semidefinite Programming for Integer Quadratic Problems*, Klagenfurt
Joint supervision with Angelika Wiegele

Master's theses

2024–2025 **Johannes Schmucker**, *Symmetric Term Sparsity*, Klagenfurt

Bachelor's theses

2026 **Eva Marktl**, *Tropical Spectrahedra*, Klagenfurt

2026 **Eleni Huber**, *The Borsuk-Ulam Theorem*, Klagenfurt

University Teaching

Lectures, exercise classes, programming classes and seminars for students of mathematics, computer science, engineering and teacher education.

University of Klagenfurt

Summer 2026 **Kombinatorische Strukturen**
Lecture and exercise class

Winter 2025/26 **Proseminar Diskrete Mathematik**

Winter 2025/26 **Diskrete Mathematik**
Lecture and exercise class

Summer 2025 **Kombinatorische Strukturen**
Lecture and exercise class

Winter 2024/25 **Diskrete Mathematik**
Lecture and exercise class

Summer 2024 **Selected Topics in Optimization: Symmetries and Semidefinite Programming**
Lecture and exercise class (course prepared from scratch)

Winter 2023/24 **Algebraische Strukturen**
Lecture and exercise class

Winter 2023/24 **Lineare Algebra 1**
Exercise class

Winter 2023/24 **Computermathematik für das Lehramt**

Winter 2023/24 **Preparatory Course for Mathematics**

Summer 2023 **Lineare Algebra 2**
Exercise class

Summer 2023 **Linear Algebra for Engineers**
Exercise class

Winter 2022/23 **Computermathematik für das Lehramt**

Winter 2022/23 **Proseminar Diskrete Mathematik**

Tilburg University

Summer 2022 **Linear Algebra for Data Science**
Tutorials

Winter 2021/22 **Linear Optimization**
Tutorials and computer labs

Academic Service

Journal refereeing *Mathematical Programming*; *Electronic Journal of Combinatorics*; *Journal of Optimization Theory and Applications*; *INFORMS Journal on Computing*; *Journal of Combinatorial Theory, Series B*.

Reviewing IPCO.

Sessions Organized sessions at EUROPT and EUCCO, including the Combinatorial Optimization focus session at EUCCO 2025.

Organization Member of the local organizing committee, EUCCO 2025, Klagenfurt.

Hiring committees Member of committees for a professorship in optimization (2024), postdoctoral positions in optimization (2023, 2025), and a doctoral position in optimization (2025).

Outreach and Third Mission

2026 **Lange Nacht der Forschung**, *University of Klagenfurt*
Lead the team for the mathematics department's station: *Wie können wir Symmetrien erkennen, beschreiben und nutzen – von der Natur bis zur abstrakten Mathematik?*

2024–2025 **Ferialpraktikum**, *University of Klagenfurt*
Helped supervise projects for secondary-school students visiting the university.

2023–2025 **Modellierungstage**, *Schools in Carinthia, Austria*
Delivered day-long workshops on mathematical topics at several schools.

Professional and Teaching Development

2026– **Aurora Outstanding**, *UiT The Arctic University of Norway*, Tromsø
Talent development program of UiT, with the aim of a succesful ERC application.

2026 **Early Career Researcher Grant Lab**, *Klagenfurt*

2024 **Erfolgreich überzeugen, argumentieren und zielgerichtet Fragen stellen**, *Klagenfurt*

2024 **Konferenzen stressfrei planen**, *Klagenfurt*

2024 **Online-Lehre interaktiv(er) gestalten mit H5P**, *Klagenfurt*

2024 **Was ist eigentlich dieses Gendern? Und wie schaut es im Uni-Alltag aus? - Ein Kurzworkshop zu gender- und diversitätssensibler Sprache**, *Klagenfurt*

2024 **Wie aktiviere ich Studierende (online)? - Einsatz von Tweedback**, *Klagenfurt*

2023 **Effective Presentation in Class and at Conferences**, *Klagenfurt*

2020 **Complementary Skills Session on intellectual property rights**, *Tilburg*

Research Visits

2025 **UiT The Arctic University of Norway**, *Tromsø, Norway*, 5 days
Visited *Cordian Riener*

2024 **Iowa State University**, *Ames, USA*, 5 days
Visited *Bernard Lidický*

2024 **Georgia Tech**, *Atlanta, USA*, 1 month
Visited *Greg Blekherman*

2024 **University of Iowa**, *Iowa City, USA*, 2 days
Visited *Kurt Anstreicher*

2023 **Georgia Tech**, *Atlanta, USA*, 10 days
Visited *Greg Blekherman*

Talks and Lectures

Invited talks and lectures

Jun 26, 2026 **Structuring Combinatorial Problems with Symmetry**, Chemnitz University of Technology, Germany
Symmetry-Reduction for Extremal Combinatorics. [slides]

Jan 7, 2026 **COW 2026**, Aussois, France
Getting to the Root of the Problem: Sums of Squares for Limits of Trees. [slides]

Oct 16, 2025 **Flag algebras and extremal combinatorics**, American Institute of Mathematics, Pasadena, USA
Derivatives of Flag Algebras (recording here)

Jul 11, 2025 **Semidefinite programming: applications & solution methods**, Lorentz Center, Leiden, the Netherlands
Optimality Conditions for Extremal Combinatorics [slides]

Jun 4, 2025 **Algebra Seminar**, UiT, Tromsø, Norway
Combinatoric Derivations and Sidorenko's Conjecture. [slides]

Mar 14, 2024 **MoPAT-24: Moments and Polynomials: Applications and Theory**, Konstanz, Germany
Combinatoric derivations in extremal graph theory and Sidorenko's conjecture. [slides]

Feb 22, 2024 **Iowa State University**, Ames, USA
New lower bounds on crossing numbers of $K_{m,n}$. [slides]

Feb 20–21, 2024 **Iowa State University**, Ames, USA
Mini Course: Symmetry reduction and semidefinite programming

Feb 13, 2024 **GT Graph Theory/Combinatorics Seminar**, Georgia Tech, Atlanta, USA
Combinatoric Derivations in Extremal Graph Theory and Sidorenko's Conjecture. [slides]

Feb 9, 2024 **Tippie College of Business**, Iowa City, USA
New lower bounds on crossing numbers of $K_{m,n}$. [slides]

Nov 13, 2023 **Symmetry, Stability, and interactions with Computation**, CIRM, Luminy, France
Flag Sums of Squares for Sidorenko's Conjecture. [slides]

Aug 17, 2023 **Mixed-integer Nonlinear Optimization: A Hatchery for Modern Mathematics**, Oberwolfach, Germany
Is the set of trees convex? [slides]

Apr 15, 2023 **Meeting on Applied Algebraic Geometry (MAAG) 2023**, Atlanta, USA
The Flag Algebra of Rooted Binary Trees. [slides]

Apr 14, 2023 **Georgia Tech Graph Theory & Combinatorics Seminar**, Atlanta, USA
New lower bounds on crossing numbers of $K_{m,n}$. [slides]

Feb 22, 2023 **Semidefinite optimization approaches to classical and quantum combinatorial optimization**, Cologne, Germany
SDPs for Extremal Combinatorics. [slides]

Dec 1, 2022 **Three days of computational methods for extremal discrete geometry**, Cologne, Germany
New lower bounds on crossing numbers of $K_{m,n}$. [slides]

Apr 12, 2022 **Workshop on Conic Linear Optimization for Computer-Assisted Proofs**, Oberwolfach
The Symmetries of Flag-Algebras. [slides]

Apr 1, 2022 **Discrete Math Seminar**, University of Massachusetts Amherst
Symmetry reduced Flag-hierarchies. [slides]

Jan 2021 **Oberseminar *Reelle Geometrie und Algebra***, Universität Konstanz
More efficient and flexible flag algebras.

Other conference and seminar presentations

Jul 2, 2026 **Cones by the Lake: Uncertain QCQPs**, Klagenfurt, Austria
Möbius-Transform Based Symmetry Reduction. [slides]

Jun 3, 2026 **SIAM Conference on Optimization (OP26)**, Edinburgh, UK
Möbius-Transform Based Symmetry Reduction. [slides]

Sep 30, 2025 **EUCCO 2025**, Klagenfurt, Austria
Optimality Conditions for Extremal Combinatorics. [slides]

Jul 23, 2025 **ICCOPT 2025**, Los Angeles, USA
Optimality Conditions for Extremal Combinatorics. [slides]

Mar 17, 2025 **KoKoKO**, Graz, Austria
Combinatoric Derivations and Sidorenko's Conjecture. [slides]

Jul 25, 2024 **ISMP 2024**, Montreal, Canada
Combinatoric Derivations and Sidorenko's Conjecture. [slides]

Jun 6, 2024 **Klagenfurt-Berlin Workshop on Multiple Perspectives in Optimization**, Klagenfurt, Austria
New lower bounds on crossing numbers of $K_{m,n}$. [slides]

May 22, 2024 **ISCO 2024: International Symposium on Combinatorial Optimization**, Tenerife, Spain
Flag Sums of Squares for Sidorenko's Conjecture. [slides]

Nov 8, 2023 **Doctoral Seminar**, Klagenfurt, Austria
Möbius Transform Based Bounds for Constant Weight Codes. [slides]

Sep 19, 2023 **ÖMG Tagung 2023**, Graz, Austria
Extremal Combinatorics in Julia. [slides]

Aug 25, 2023 **EUROPT 2023**, Budapest, Hungary
The flag algebra of rooted binary trees. [slides]

Jul 10, 2023 **SIAM Conference on Applied Algebraic Geometry (AG23)**, Eindhoven, the Netherlands
Möbius Transform Based Bounds for Constant Weight Codes. [slides]

Jun 2, 2023 **SIAM Conference on Optimization (OP23)**, Seattle, USA
Flag Sums of Squares for Sidorenko's Conjecture. [slides]

Oct 23, 2022 **University of Klagenfurt**, Klagenfurt, Austria
Derivatives in Continuous Combinatorics. [slides]

Jul 26, 2022 **ICCOPT**, Bethlehem, PA, USA
Möbius-Transform Based Symmetry Reduction for Optimization in Binary Variables. [slides]

Mar 23, 2022 **Polynomial optimization reading group**, CWI, Amsterdam
Symmetry reduced Flag-hierarchies.

Aug 20, 2021 **SIAM AG21**
More efficient and flexible Flag-Algebras coming from polynomial optimization.

Jul 20–23, 2021 **SIAM OP21**
More efficient and flexible Flag-Algebras coming from polynomial optimization.

Feb 2021 **Virtual OR seminar**, Tilburg University
More efficient and flexible Flag-Algebras.

Jan 2021 **Shared seminar Cologne Oberseminar/CWI reading group**
More efficient and flexible Flag-Algebras.

Feb 26, 2020 and Mar 4, 2020 **Polynomial optimization reading group**, CWI, Amsterdam
A two-part introduction to symmetry reduction for SDPs

Aug 7, 2019 **ICCOPT**, Berlin
Minimum energy configurations on a toric lattice as a quadratic assignment problem.

Conferences, Workshops and Research Training

Sep 13–18, 2026 **13th Heidelberg Laureate Forum**, Heidelberg, Germany
(Competitive application procedure.)

Sep 9–11, 2026 **Nasjonalt matematikermøte 2026**, Trondheim, Norway

Jul 1–3, 2026 **Cones by the Lake: Uncertain QCQPs**, Klagenfurt, Austria

Jun 24–26, 2026 **Structuring Combinatorial Problems with Symmetry**, Chemnitz, Germany

Jun 2–5, 2026 **SIAM Conference on Optimization (OP26)**, Edinburgh, UK

Jan 5–9, 2026 **COW 2026**, Aussois, France

Oct 13–17, 2025 **AIM workshop: Flag algebras and extremal combinatorics**, Pasadena, USA

Sep 29–Oct 1, 2025 **EUCCO 2025**, Klagenfurt, Austria

Jul 19–24, 2025 **ICCOPT 2025**, Los Angeles, USA

Jul 7–11, 2025 **Semidefinite programming: applications & solution methods**, Lorentz Center, Leiden, the Netherlands

Mar 17, 2025 **KoKoKo**, Graz, Austria

Jul 21–26, 2024 **ISMP 2024**, Montreal, Canada

Jun 6–7, 2024 **Klagenfurt-Berlin Workshop on Multiple Perspectives in Optimization**, Klagenfurt, Austria

May 22–24, 2024 **ISCO 2024: International Symposium on Combinatorial Optimization**, Tenerife, Spain

Mar 11–14, 2024 **MoPAT-24: Moments and Polynomials: Applications and Theory**, Konstanz, Germany

Nov 13–17, 2023 **Symmetry, Stability, and interactions with Computation**, CIRM, Luminy, France

Sep 18–22, 2023 **ÖMG Tagung 2023**, Graz, Austria

Aug 23–25, 2023 **EUROPT 2023**, Budapest, Hungary

Aug 13–18, 2023 **Mixed-integer Nonlinear Optimization: A Hatchery for Modern Mathematics**, Oberwolfach, Germany

Jul 10–14, 2023 **SIAM Conference on Applied Algebraic Geometry (AG23)**, Eindhoven, the Netherlands

May 31–Jun 3, 2023 **SIAM Conference on Optimization (OP23)**, Seattle, USA

Apr 15–16, 2023 **Meeting on Applied Algebraic Geometry (MAAG) 2023**, Atlanta, USA

Feb 22–23, 2023 **Semidefinite optimization approaches to classical and quantum combinatorial optimization**, Cologne

Nov 30–Dec 2, 2022 **Three days of computational methods for extremal discrete geometry**, Cologne

Sep 5–9, 2022 **Final POEMA workshop**, Paris

Jul 23–28, 2022 **ICCOPT**, Bethlehem, PA, USA

Jun 7–9, 2022	**Nordic Combinatorial Conference (NORCOM)**, Tromsø
Apr 10–15, 2022	**Workshop on Conic Linear Optimization for Computer-Assisted Proofs**, Oberwolfach
Jun 21–29, 2021	**MINOA Doctoral School 2021**, Online
Apr 16, 2021	**General Julia training (POEMA)**, Online
Mar 4–5, 2021	**Second MINOA ESR days**, Online
Mar 1–3, 2021	**Annual MINOA Conference 2021**, Online
Jan—Mar 2021	**POEMA 3rd Workshop**, Online
Nov 23–24, 2020	**First MINOA ESR days**, Online
Oct–Dec 2020	**POEMA 2nd Workshop**, Online
May 27–Sep 16, 2020	**POEMA Online Learning Weeks**, Online
Jan 6–10, 2020	**2nd MINOA conference**, Aussois, France
Jan 6–10, 2020	**24th Workshop on Combinatorial Optimization**, Aussois, France
Sep 9–Nov 11, 2019	**Interior Point Methods**, *LNMB PhD Course*, Etienne de Klerk, Utrecht, the Netherlands
Aug 5–8, 2019	**6th International Conference on Continuous Optimization (ICCOPT)**, Berlin, Germany
Jun 24–28, 2019	**1st MINOA PhD school**, *Mixed-Integer Nonlinear Optimization meets Data Science* , Ischia, Italy
Jan 14–16, 2019	**1st MINOA conference**, Aussois, France
Jan 14–16, 2019	**23rd Workshop on Combinatorial Optimization**, Aussois, France
Jan 7–11, 2019	**44th conference on the mathematics of operations research**, Lunteren, the Netherlands
Nov 19–Feb 18, 2019	**Networks and Semidefinite Programming**, *LNMB PhD Course*, Monique Laurent, Utrecht, the Netherlands
2019–2022	**CWI reading group on polynomial optimization**, *hosted by* Monique Laurent *and* Sven Polak, CWI, Amsterdam
2020–2022	**Oberseminar**, *hosted by* Frank Vallentin, Cologne

Languages

German	Native
English	C2
Dutch	B1

Personal Details

Date of birth	November 18, 1996
Place of birth	Leverkusen, Germany
Citizenship	German

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